

Student Name: Kate Hsu

Student ID#: 20956017

ECON 332 – INTERNATIONAL FINANCE
MIDTERM EXAM #2
EXAMINER: Muhebullah Karimzada

March 19, 2025

Instructions: You have 75 minutes to work on the exam (from 2:30 pm to 3:45 pm). This midterm is individual and closed-book. For the multiple choice questions, choose the option that best answers/completes the question/enunciate, and transfer your answer to the front page.

PLEASE: PUT YOUR NAME AND STUDENT ID ON EVERY SHEET!!

Multiple Choice Answers:

1. C

9. B

17. B

2. A

10. A

18. C

3. A

11. A

19. D

4. D

12. A

20. A

5. A

13. C

6. D

14. A

7. A

15. C

8. D

16. B

~~36~~
60

PART I – MULTIPLE CHOICE QUESTIONS (2.5 points each question)

1. Summaries of international flows of goods and assets are recorded in a nation's:

- A) national income and product accounts.
- B) balances with the IMF.
- ☒ C) balance of payments accounts.
- D) balance on international indebtedness.

2. Income paid to factors is called:

- ☒ A) national income.
- B) value added.
- C) net exports.
- D) the current account.

3. The total economic activity in a nation is an important measure. There are three approaches that economists use to measure key indicators. Which is a method that economists do NOT use to measure economic activity?

- ☒ A) the accounting approach, using the idea that total private sales have to equal total private production
- B) the income approach measuring income received by factors of production
- C) the expenditure approach, using the idea that the total economic activity is equal to the combined purchases of the various sectors
- D) the product approach, which measures GDP from a production standpoint

4. How do we handle intermediate goods and services when converting from GNE to GDP?

- ☒ A) Intermediate goods are already counted in GNE, so they do not have to be added in to GDP.
- B) Imported intermediate goods are counted in GNE, so they must be subtracted from GDP to avoid double counting.
- C) Exported intermediate goods are counted in GNE, so they must be subtracted from GDP to avoid double counting.
- D) We add (to GNE) the value of imported goods and services and subtract the value of exported goods and services to get GDP.

5. In addition to summarizing international flows of goods and services, factor inputs, and transfers, ____ includes financial flows such as deposits, purchases of stocks, bonds investment in plant and equipment, and real estate.

- ☒ A) the financial account
- B) balance of payments accounting
- C) the current account
- D) the capital account

Use the following table to answer questions 6-7.

Table: Hypothetical Canadian National Income and Product Accounts Data

Category	Billions of dollars
Consumption (personal consumption expenditures)	8,000
Investment (gross private domestic investment)	1,300
Government consumption (government expenditures)	2,100
Exports	900
Imports	1,750
NFIA	+45
Net unilateral transfers	+20

6. According to the NUTs, economists would say that:

- A) Canada has given away more foreign aid than it receives.
- B) Canadians abroad have sent a lot of income home.
- C) Mexicans in Canada have sent a lot of income home.
- ☒ D) Canada has a balance of payments deficit.

7. GNDI for the economy provided is:

- ☒ A) \$11,825 billion.
- B) \$10,615 billion.
- C) \$8,625 billion. ✗
- D) \$8,600 billion. ✗

$$C + I + G + 20$$

8. Private savings deficits plus government budget deficits contribute to higher current account deficits. Some economists refer to the Ricardian equivalence theory to assert that;

- A) the public will offset its future tax liability to some degree by increasing saving.
- B) the public abhors private and public debt and will demand an end to deficits.
- C) savings will increase as the population ages, since older people save more.
- ☒ D) the current account deficit is caused by trade and is not influenced by deficit spending.

9. Whenever there is an outflow of funds from any of the balance of payments (BOP) accounts, it is recorded as a _____, and called a _____.

- A) minus; unilateral transfer
- ☒ B) minus; BOP debit
- C) plus; BOP credit
- D) plus; current account surplus

10. Suppose that a Japanese investor buys a Korean stock with cash. For Korea, this counts as a:

- ☒ A) current account credit, current account debit. ~~X~~
- ☐ B) financial account debit, financial account credit. ~~X~~
- ☐ C) capital account debit, capital account credit. ~~X~~
- ☐ D) financial account debit, current account debit.

get money
→ export +10
credit CA ↑

11. Assume a nation's external wealth is negative. Also assume all liabilities and assets are denominated in a foreign currency. How will its wealth change when its currency appreciates in world markets?

- ☒ A) Its external wealth will rise.
- ☐ B) Its external wealth will fall.
- ☐ C) It depends on the currency in which it has its assets and liabilities.
- ☐ D) There will be no change in the value of its wealth.

$\$1 = \$1.5 \Rightarrow \$1 = \1.1
stronger

12. Which of the following is NOT an assumption belonging to the long-run budget constraint model?

- ☒ A) The government has a balanced budget.
- ☐ B) Prices are perfectly flexible.
- ☐ C) The economy is a price taker, small, and open. ✓
- ☐ D) The economy can borrow or lend unlimited amounts at the world real interest rate. ✓

spend what you make

13. Which of the following is the value of a country's external wealth at the end of 2016 if its external wealth was \$1 billion at the end of 2015 and its trade balance was \$500 million in 2016? Assume the world interest rate is 10% per annum.

- ☐ A) \$1.5 billion
- ☐ B) \$1.6 billion
- ☒ C) \$0.5 billion
- ☐ D) \$0.6 billion

TB = 500
 $i^* = 10\%$

$A = 1$

14. The two-period budget constraint tells us:

- ☒ A) a creditor country can afford future deficits "on average."
- ☐ B) a creditor country cannot afford future deficits "on average."
- ☐ C) a debtor country cannot afford future deficits "on average."
- ☐ D) a debtor country can afford future deficits "on average."

$A = 0$

$B = 1$

$B = 0$

15. The long-run budget constraint dictates that:

- ☐ A) the present value of trade balances is zero taken to infinity. ~~X~~
- ☐ B) the present value of wealth is equal to zero taken to infinity. ~~X~~
- ☒ C) the present value of wealth is exactly offset by the present value of trade balances.
- ☐ D) if the present value of wealth is positive, the nation cannot afford future trade deficits on average.

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Use the following to answer questions 16-18.

SCENARIO: IS-LM-FX

Assume that initially the IS curve is given by

$$IS_1: Y = 12 - 1.5T - 30i + 2G,$$

Assume as well that the price level P is 1, and the LM curve is given by

$$LM_1: M = Y(1 - i).$$

And that the forex market equilibrium is characterized by:

$$i = ([E^e / E] - 1) + 0.10$$

Finally, assume that $T=2$, $G=1.5$, and the home interest rate is 0.2 (i.e. 20%).

16. (Scenario: IS-LM-FX) What is the level of output according to the IS_1 curve?

- A) 5.5
- ☒ B) 6
- C) 9
- D) 10

$$Y = 12 - 1.5(2) - 30(0.2) + 2(1.5)$$

17. (Scenario: IS-LM-FX) If the level of output is the one determined by the IS_1 curve, what is the equilibrium level of home money supply?

- A) 3.5
- ☒ B) 4.8
- C) 5.2
- D) 9

$$6(1 - 0.2)$$

18. (Scenario: IS-LM-FX) If the expected exchange rate is equal to 1, how much should the current value of the spot exchange rate be?

- A) 0.5
- B) 0.81
- ☒ C) 0.91
- D) 1

$$0.2 = \left[\frac{1}{E} - 1 \right] + 0.1$$

$$\cancel{1.1} = \frac{1}{E} \quad 1/E = 1$$

$$= C + C(m) \quad \uparrow \text{MPK}$$

19. The TB (i.e., $X - M$) is part of the short-run spending equation. With sticky prices, what would be the effect on the TB with an increase (a real depreciation) of the home nation's exchange rate?

- more foreign goods
→ demand increase
→ exports increase
- get wealth, spend less
- A) Consumers in the home nation would find it more expensive to buy domestic goods compared with foreign goods, and the TB would decrease.
- B) Consumers in the home nation would cut back on both domestic and foreign goods and the TB would decrease.
- C) Consumers in the home nation would increase spending on both domestic and foreign goods, and the TB would be unchanged.
- D) Consumers in the home nation would increase spending on domestic goods and decrease spending on foreign goods, causing the TB to increase.

20. If the trade surplus has fallen, which of the following is a possible explanation?

- A) The real exchange rate rose.
- B) Foreign income fell.
- C) Domestic income fell.
- D) The foreign price level rose.

$$TB = EX - IM$$

PART II – PROBLEM (40 points in total. 5 points each sub-question)

In this question assume all dollar units are real dollars in billions, so \$150 means \$150 billion. It is year 0. Argentina thinks it can find \$150 of domestic investment projects with an MPK of 20% (each \$1 invested pays off \$0.20 in every later year). Argentina invests \$96 in year 0 by borrowing \$96 from the rest of the world at a world real interest rate r^* of 10%. There is no further borrowing or investment after this.

Use the standard assumptions: Assume initial external wealth W (W in year -1) is 0. Assume $G=0$ always; and assume $I=0$ except in year 0. Assume $NUT = KA = 0$ and that there is no net labor income so that $NFIA = r^* W$.

The projects start to pay off in year 1 and continue to pay off all years thereafter. Interest is paid in perpetuity, in year 1 and every year thereafter. In addition, assume that if the projects are *not* done, then $GDP = Q = C = \$300$ in all years, so that: $PV(Q) = PV(C) = 300 + 300/0.10 = 3,300$.

- a) Should Argentina fund the \$96 worth of projects? Why or why not? Explain your answer.

$$301.6$$

$$300$$

$$319.2$$

$$300$$

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$$1.6 / 0.04$$

- b) From this point forward, assume the projects totaling \$96 are funded and completed in year 0. If the MPK is 20%, what is the total payoff from the projects in future years?
- c) Assume this is added to the \$300 of GDP in all years starting in year 1. In $300 + 1.6$ dollars, what is Argentina's $Q = GDP$ in year 0, year 1, and later years?
- d) At year 0, what is the new $PV(Q)$ in dollars? Hint: To simplify, calculate the value of the increment in $PV(Q)$ due to the extra output in later years. $q + \frac{q}{r}$
- e) At year 0, what is the new $PV(I)$ in dollars? Therefore, what does the LRBC say is the new $PV(C)$ in dollars? $PV(q) - PV(I)$
- f) Assume that Argentina is smoothing consumption. What is the percent change in $PV(C)$? What is the new level of C in all years? Is Argentina better off? $\frac{a-b}{b}$
- g) For the year the investment projects go ahead, year 0, explain Argentina's balance of payments as follows: State the levels of CA , TB , $NFIA$, and FA . $C + \frac{C}{r^*}$
- h) What happens in later years? State the levels of CA , TB , $NFIA$, and FA in year 1 and every later year.

a) $MPK > r^* \Rightarrow 0.2 > 0.1$ Since MPK is lower than r^* , Argentina should fund the project

b) $\$96 B \times 0.2 = \19.2 $PV(\text{payoff}) = \frac{19.2}{0.1} = \192

the total payoff is \$192 B

c) Year 0: \$300 GDP

Year 1 = Year 2 = ... = $300 + 19.2 = \$319.2$ GDP

in year 0, Argentina's $Q = GDP = \$300 B$. After year 0, the years 1 to later years will be equivalent, \$319.2 B.

$$C \left[1 + \frac{1}{2} \right] 1.5C = 5$$

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Answers to the Problem (please be clear!!)

d) $PV(Q) = q + \frac{q_0}{r^*} = 319.2 + \frac{300}{0.1} = \$3,319.2$

the new $PV(Q)$ is \$3,319.2

e) $PV(Q) - PV(I) = 3319.2 - 3300 = \19.2

we assumed $w(w \text{ in year } -1) = 0 \Rightarrow NFA = r^*w = 0$

the new $PV(I) = 0$ since $NFA = r^*w$.

the CRBC says $PV(I) = \$19.2$

f) % change $\Delta PV = \frac{\text{after} - \text{before}}{\text{before}} = \frac{3319.2 - 3300}{3300} = 0.05818 = 5.82\%$

$$PV(C) = C + \frac{C}{r^*}$$

$$C = 30174.55$$

$$3319.2 = C + \frac{C}{0.1}$$

The percentage change is 5.82% and new level of C is 30174.55. Yes, they are better off.

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Answers to the Problem (please be clear!!)

	Year 0	n> Year 1, 2, ...
g) G	0	0
L	0	0
CA	300	300
TB	300	319.2
NFIA	$0.1 \times 300 = 30$	$0.1 \times 319.2 = 31.92$
FA	300	319.2

$$NFIA = r^* W$$

$$NUT = KA = 0$$

$$C_1 = \frac{C}{r^*}$$

$$G = 0$$

$$I = 0 \text{ not in yr 0}$$

$$TB = \text{Exports} - \text{Imports}$$

$$CA = FA -$$

$$CA$$

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Answers to the Problem (please be clear!!)